THE EXACT ZARANKIEWICZ NUMBER $\mathrm{Z}(13, 22, 3, 3)$

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Abstract. The Zarankiewicz number $\mathrm{Z}(m, n, s, t)$ is the maximum number of edges in a bipartite graph with parts of orders m and n containing no copy of $K_{s,t}$. Recent construction search established a 137-edge $K_{3,3}$-free graph on parts of orders 13 and 22, while the best published upper bound recorded for this parameter was 140. We prove that no 138-edge example exists and hence determine the exact value

$$\mathrm{Z}(13, 22, 3, 3) = 137.$$

The upper bound is a finite, certificate-based computer-assisted proof. Triple counting and a supporting-line penalty reduce a hypothetical 138-edge graph to 83 column-degree profiles. Exact rational block-polytope certificates exclude 77 profiles, and one further profile is removed by an elementary marked-row congruence argument. For each of the five remaining profiles, a marked row produces a twelve-point residual packing and a leave multigraph. Point-type equations, modular Gram-rank tests, determinant-square conditions, exhaustive leave enumeration, and exact Farkas certificates eliminate every residual case. The accompanying verifier uses integer and rational arithmetic only; floating-point optimization is used solely to discover certificates and is not part of the proof trust boundary.

1. Introduction

The Zarankiewicz problem asks for the maximum number of edges in a bipartite graph avoiding a specified complete bipartite subgraph. It was posed by Zarankiewicz in 1951 [7] and was placed in its modern extremal form by Kővári, Sós and Turán [4]. Although the asymptotic theory is extensive, exact values for moderate finite parameters remain difficult. The difficulty is particularly pronounced for $K_{3,3}$-free graphs: useful counting inequalities leave a small but highly structured set of integer configurations, while direct exhaustive search suffers from severe symmetry and proof-certification costs.

Write $\mathrm{Z}(m, n, s, t)$ for the maximum number of edges in a subgraph of $K_{m,n}$ containing no $K_{s,t}$. Roman's linear-programming framework [5] and later strengthened relaxations [2] provide effective finite upper bounds. SAT-based methods have also been developed for exact finite cases [6]. On the lower-bound side, Bhan, Nobili and Langer [1] recently obtained a 137-edge construction for the present parameter, while the prior upper bound was 140 [2]. Thus

$$137 \leq \mathrm{Z}(13, 22, 3, 3) \leq 140.$$

Their work also determined three neighboring exact values, including $\mathrm{Z}(12, 22, 3, 3) = 132$.

The contribution of this paper is the matching upper bound.

Theorem 1.1. *There is no 13×22 binary matrix with 138 ones and no all-one 3×3 submatrix. Consequently,*

$$\boxed{\mathrm{Z}(13, 22, 3, 3) = 137}.$$

The proof is computer-assisted but certificate based. It is not a report of a mixed-integer solver returning "infeasible." Every load-bearing numerical assertion is replayed by short exact-arithmetic programs from explicit finite data. The architecture has three layers.

Date: 1 August 2026.

2020 *Mathematics Subject Classification.* 05C35, 05D99, 68R10, 90C05.

Key words and phrases. Zarankiewicz number, extremal bipartite graph, forbidden submatrix, computer-assisted proof, Farkas certificate, block design.

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(i) A global degree-profile argument reduces 138 edges to 83 profiles and eliminates 77 by exact rational Farkas certificates.
(ii) A direct marked-row congruence argument excludes the exceptional profile $6^{18}7^{3}9^{1}$.
(iii) The remaining five profiles are reduced to finite residual block-design problems on twelve points. Exhaustive leave-multigraph enumeration, necessary Gram conditions, and exact Farkas certificates exclude all residual cases.

The accompanying repository includes an explicit 137-edge witness, all certificates, the exhaustive enumerator, independent profile enumerators, and a one-command verification gate, so the computation can be independently reproduced.

2. Block formulation and the lower bound

Let $X = [13]$. Identify the jth column of a binary 13×22 matrix with its support $E_j \subseteq X$, and put $d_j = |E_j|$. For each triple $T \in \binom{X}{3}$, define

$$\lambda_T = |\{j : T \subseteq E_j\}|.$$

The matrix is $K_{3,3}$-free if and only if

$$\lambda_T \le 2 \qquad (T \in \binom{X}{3}). \tag{1}$$

Repeated columns are allowed throughout.

The supplementary file `proof/data/z13_22_137_blocks.json` contains 22 blocks with total size 137. Direct enumeration of all $\binom{13}{3} = 286$ triples verifies (1). Thus

$$\mathrm{Z}(13, 22, 3, 3) \ge 137. \tag{2}$$

The same lower bound was first reported in [1]. A matrix plot of the independently checked witness is shown in Figure 1; its block supports are listed in Appendix A.

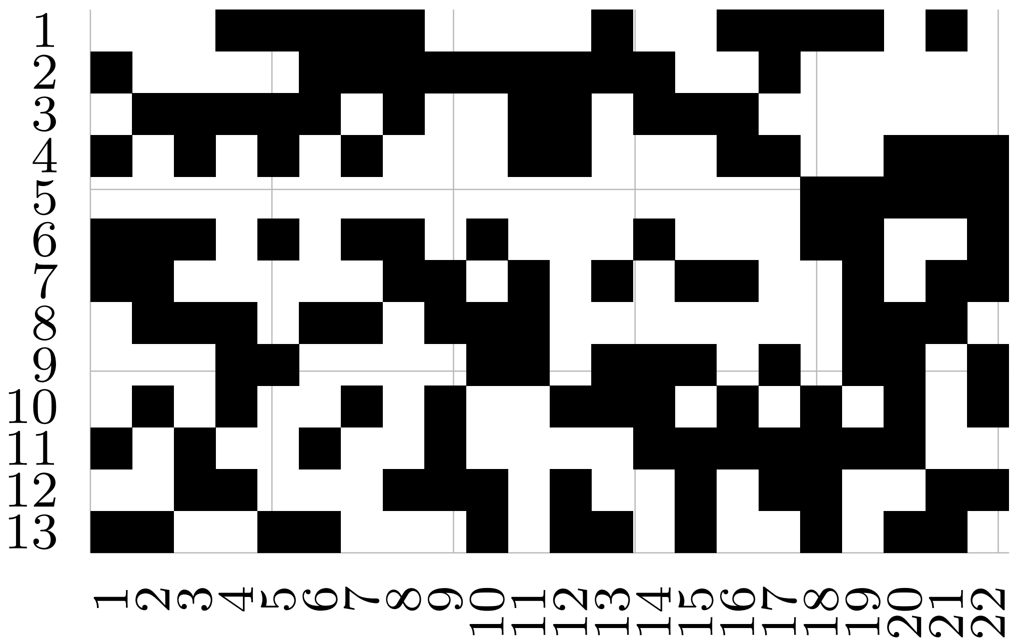

Figure 1. A checked 13×22 matrix with 137 ones. Black cells are ones. Every triple of rows is contained in at most two columns.

3. Global reduction at 138 edges

Assume for contradiction that blocks $E_1, \dots, E_{22}$ satisfy (1) and

$$\sum_{j=1}^{22} d_j = 138. \tag{3}$$

Double counting incidences between blocks and row triples gives

$$\sum_{j=1}^{22} \binom{d_j}{3} = \sum_{T \in \binom{X}{3}} \lambda_T \le 2\binom{13}{3} = 572. \tag{4}$$

Define the integer penalty

$$p(d) = \binom{d}{3} - 15d + 70, \qquad 0 \le d \le 13. \tag{5}$$

Its values are

d	0	1	2	3	4	5	6	7	8	9	10	11	12	13
$p(d)$	70	55	40	26	14	5	0	0	6	19	40	70	110	161.

Thus $p(d) \ge 0$, with equality precisely for $d = 6, 7$. Combining (3)–(5),

$$\sum_{j=1}^{22} p(d_j) = \sum_j \binom{d_j}{3} - 15 \cdot 138 + 70 \cdot 22 \le 42. \tag{6}$$

Exact integer enumeration of all histograms $(n_0, \dots, n_{13})$ satisfying

$$\sum_d n_d = 22, \qquad \sum_d d n_d = 138, \qquad \sum_d p(d) n_d \le 42$$

leaves exactly 83 degree profiles. This is independently regenerated by `enumerate138.py` and `verify_138_reduction.py`; no degree window is assumed.

3.1. **The block-polytope relaxation.** For a profile (n_d), choose a degree f with $n_f > 0$ and, by row symmetry, fix one f-block as $F = \{1, \dots, f\}$. For every remaining allowed support $B \subseteq X$, let $x_B \ge 0$ denote its multiplicity. Every actual matrix yields an integral feasible point of the following relaxation:

$$\sum_{B \supseteq T} x_B \le 2 - \mathbf{1}[T \subseteq F] \qquad \left(T \in \binom{X}{3}\right), \tag{7}$$

$$\sum_{B \ni r} \binom{|B|-1}{2} x_B \le 132 - \mathbf{1}[r \in F]\binom{f-1}{2} \qquad (r \in X), \tag{8}$$

$$\sum_{B \supseteq P} (|B|-2) x_B \le 22 - \mathbf{1}[P \subseteq F](f-2) \qquad \left(P \in \binom{X}{2}\right), \tag{9}$$

$$\sum_{|B|=d} x_B = n_d - \mathbf{1}[d = f] \qquad (d \text{ active}). \tag{10}$$

The row and pair inequalities are nonnegative sums of triple-capacity inequalities; they are retained because they yield much shorter dual certificates.

Lemma 3.1 (Exact Farkas criterion)**.** *Write (7)–(9) as $Ax \le b$, and (10) as $Ex = e$, with $x \ge 0$. If rational vectors $\alpha \ge 0$ and unrestricted β satisfy*

$$A^T \alpha + E^T \beta \ge 0, \qquad b^T \alpha + e^T \beta < 0,$$

then the profile is impossible.

Proof. For a feasible x,

$$0 \le x^T(A^T\alpha + E^T\beta) = \alpha^T Ax + \beta^T Ex \le \alpha^T b + \beta^T e < 0,$$

a contradiction. This is the standard alternative theorem of Farkas [3]. $\square$

The verifier regenerates every subset B of each active degree and checks every dual coefficient using `fractions.Fraction`. Stored rational certificates exclude 77 of the 83 profiles. The six not separated by this global relaxation are

$$6^{18}7^39^1, \quad 6^{16}7^6, \quad 5^16^{14}7^7, \quad 5^26^{12}7^8, \quad 4^16^{13}7^8, \quad 5^36^{10}7^9. \tag{11}$$

4. Marked-row deficits

For every triple T, put

$$\delta_T = 2 - \lambda_T \ge 0, \qquad D_r = \sum_{T \ni r} \delta_T.$$

If

$$s = 572 - \sum_j \binom{d_j}{3}$$

is the unused triple capacity, then

$$\sum_{r \in X} D_r = 3s. \tag{12}$$

Since a fixed row belongs to $\binom{12}{2} = 66$ triples, each of capacity two,

$$D_r = 132 - \sum_{j : r \in E_j} \binom{d_j - 1}{2}. \tag{13}$$

This formula gives both congruence restrictions and a local design interpretation.

4.1. **The profile $6^{18}7^39^1$.** For this profile, $s = 23$, so $\sum_r D_r = 69$. Contributions from degree-six and degree-seven blocks in (13) are 10 and 15, both zero modulo five; the degree-nine contribution is $28 \equiv 3 \pmod 5$. Thus

$$D_r \equiv \begin{cases} 2 \pmod 5, & r \notin E_9, \\ 4 \pmod 5, & r \in E_9. \end{cases}$$

The residue-minimal total is $4 \cdot 2 + 9 \cdot 4 = 44$. Since the actual total is 69, only five increments of size five are available, and at least eight rows have residue-minimal deficit.

For a row outside the degree-nine block with $D_r = 2$, if a, b count degree-six and degree-seven blocks through r, then (13) gives $2a + 3b = 26$, hence $(a, b) = (13, 0)$ or $(10, 2)$. Deleting r leaves blocks of sizes five and six on twelve points. Pair-capacity degrees imply, respectively, total size-five incidence at most $60 < 65$, or at most $48 < 50$. Both are impossible. Therefore all four rows outside E_9 consume an increment.

At least eight rows inside E_9 consequently have $D_r = 4$. For such a row, $2a + 3b = 20$. The case $(a, b) = (10, 0)$ is ruled out by the residual degree-nine block: its eight points permit at most three size-five incidences and the other four at most five, totaling $44 < 50$. Hence each of these at least eight rows lies in exactly two of the three degree-seven columns. Yet any fixed pair of degree-seven columns can share at most two such rows inside E_9, since three shared rows together with E_9 would form a $K_{3,3}$. The three pairs account for at most six rows, a contradiction.

Proposition 4.1. *The profile $6^{18}7^39^1$ is impossible.*

5. The marked-row leave method

It remains to exclude the five profiles in (11) containing only degrees four through seven. Fix a row r and delete it from every block containing it. The resulting residual blocks lie on the twelve-point set $X \setminus \{r\}$.

Definition 5.1. *The* leave multigraph L_r *has vertex set* $X \setminus \{r\}$ *and gives the pair* $\{x, y\}$ *multiplicity* δ_{rxy}. *Its total edge multiplicity is* D_r *and every edge multiplicity lies in* $\{0, 1, 2\}$.

Suppose the residual blocks have active sizes s and that point x occurs in $u_{x,s}$ blocks of size s. Pair capacity through $\{r, x\}$ gives

$$e_x = 22 - \sum_s (s-1)u_{x,s}, \tag{14}$$

where e_x is the degree of x in L_r. In addition,

$$\sum_x u_{x,s} = sn_s, \qquad \sum_x e_x = 2D_r. \tag{15}$$

Equations (14)–(15) give a finite list of point-type multisets.

Let M be the point-by-residual-block incidence matrix. Its Gram matrix $Q = MM^T$ is determined by the point types and the leave:

$$Q_{xx} = \sum_s u_{x,s}, \qquad Q_{xy} = 2 - \delta_{rxy} \quad (x \neq y). \tag{16}$$

Consequently,

$$\operatorname{rank}_{\mathbb{F}_p}(Q) \leq q \tag{17}$$

for every prime p, where q is the number of residual blocks. When $q = 12$, M is square and

$$\det(Q) = \det(M)^2 \tag{18}$$

must be a nonnegative integer square. The verifier uses (17) for $p = 101, 103, 107$ and (18) only as necessary conditions. Cases passing these screens are retained.

For every surviving typed leave, the possible residual blocks form a finite linear factorization system. It requires the prescribed incidence count of every point in every size class, the prescribed pair multiplicities $2 - \delta_{rxy}$, and the prescribed number of blocks of each size. An integral residual design would give a nonnegative integral solution. Each screened case is instead excluded by an integer-scaled rational Farkas vector whose inequalities are checked over every eligible residual block.

Proposition 5.2 (Certified local screen)**.** *For each local parameter tuple invoked below, the program* `local_screen_general.cpp` *enumerates every point-type distribution satisfying* (14)–(15)*, every loopless leave multigraph with edge multiplicities at most two and the prescribed degree sequence, and every case surviving* (17) *or* (18)*. The Python verifier then checks an exact Farkas certificate for every retained factorization case.*

The proposition is a finite computational statement. Its implementation is deliberately small: the C++ source is approximately four kilobytes, and the certificate checker uses only the Python standard library. The complete enumeration counts used below are summarized in Table 1.

6. The common deficit-seven branch

Four of the remaining profiles may force a row outside all degree-five blocks with $D_r = 7$. If a, b denote the numbers of degree-six and degree-seven columns through r, then (13) leaves

$$(a, b) = (5, 5), (8, 3), (11, 1).$$

The $(11, 1)$ branch has no point-type distribution. For $(5, 5)$, all 8,641 leave graphs fail the Gram-rank condition. The $(8, 3)$ branch has 37 point-type distributions and 6,733 leave

Branch	Local parameters	type multisets	leaves	certified survivors
Common $D = 7$	$(a, b) = (5, 5)$	19	8,641	0
Common $D = 7$	$(a, b) = (8, 3)$	37	6,733	4
$5^1 6^{14} 7^7$, inside	$(a, b) = (6, 4)$	508	43,715	0
$5^1 6^{14} 7^7$, inside	$(a, b) = (3, 6)$	6	534	0
$5^2 6^{12} 7^8$, $c = 2$	$(10, 1), (7, 3), (4, 5)$	–	–	195+217+28
$5^3 6^{10} 7^9$, $c = 3$	$(8, 2), (5, 4), (2, 6)$	–	–	4,746+656+3
$4^1 6^{13} 7^8$, inside	$(8, 3), (5, 5)$	41+48	367+275	12+3

TABLE 1. Selected exact local-enumeration counts. A dash indicates that only the final screened-case count is used in the proof report.

graphs; four cases survive the Gram screen. Three have immediate exact local Farkas certificates.

The fourth case is fractionally feasible and therefore requires an integral refinement. The verifier enumerates all systems of its three residual size-six blocks, obtaining exactly 285 systems. The full automorphism group preserving the typed leave is $S_2 \times S_3 \times S_6$, of order 8,640, and partitions the systems into three orbits of sizes 15, 180, and 90. The first two representatives have no completion by eight residual size-five blocks; the third has exactly four completions. For each of these four local completions, and for each of the four relevant global profiles, a separate rational completion certificate excludes the remaining columns. Thus sixteen exact completion certificates close the branch.

Lemma 6.1. *No marked row outside all exceptional small blocks can realize the common deficit-seven configuration used in the profiles* $6^{16}7^6$, $5^1 6^{14} 7^7$, $5^2 6^{12} 7^8$, *or* $5^3 6^{10} 7^9$.

7. Elimination of the five final profiles

We now combine residue baselines with Proposition 5.2 and Lemma 6.1.

7.1. The profile $6^{16}7^6$. Here $s = 42$, so $\sum_r D_r = 126$, and every $D_r \equiv 2 \pmod 5$. If no row had deficit 2 or 7, then every deficit would be at least 12, giving a total at least 156. Every deficit-two local type is rejected by point-type or Gram screening, while every deficit-seven type is rejected by Lemma 6.1. Hence the profile is impossible.

7.2. The profile $5^1 6^{14} 7^7$. Let $c \in \{0, 1\}$ record membership in the unique degree-five block. Then $D_r \equiv 2 - c \pmod 5$. The minimum cases $D = 2$ for $c = 0$ and $D = 1$ for $c = 1$ are locally impossible. The next baseline has deficit seven on the eight outside rows and six on the five inside rows, totaling

$$8 \cdot 7 + 5 \cdot 6 = 86.$$

The required total is 111, only five increments of size five larger, so some row attains this baseline. Outside rows are impossible by Lemma 6.1; the inside possibilities

$$(a, b) = (3, 6), (6, 4), (9, 2), (12, 0)$$

are all rejected by exact local screening. Thus the profile is impossible.

7.3. The profile $5^2 6^{12} 7^8$. Let $c \in \{0, 1, 2\}$ count the two degree-five blocks through the marked row. The initial cases $(c, D) = (0, 2), (1, 1), (2, 0)$ are impossible. The next residue baseline has total

$$7 \cdot 13 - 10 = 81,$$

whereas the required total is 96. Thus a baseline row exists. The $c = 0$ and $c = 1$ branches reduce to earlier screens. For $c = 2, D = 5$, the possible values are

$$(a, b) = (10, 1), (7, 3), (4, 5), (1, 7).$$

The last has no point types. The first three yield 195, 217, and 28 screened cases, respectively, all excluded by exact Farkas certificates.

7.4. **The profile $5^3 6^{10} 7^9$.** The initial $c = 0, 1, 2$ cases are again impossible. The next baseline has total

$$7 \cdot 13 - 15 = 76,$$

while the required total is 81, so a baseline row exists. The branches $c = 0, 1, 2$ have already been excluded. For $c = 3, D = 4$, the possibilities

$$(a, b) = (8, 2), (5, 4), (2, 6)$$

produce 4,746, 656, and 3 screened local cases. Every one of the 5,405 factorization systems has an exact integer-scaled Farkas certificate.

7.5. **The profile $4^1 6^{13} 7^8$.** There are nine rows outside and four inside the degree-four block. Their minimum-residue baseline is

$$9 \cdot 2 + 4 \cdot 4 = 34,$$

whereas the required total is 84. If neither minimum occurred, the total would be at least 99, so a row of deficit two outside or deficit four inside must occur. Outside deficit-two rows are impossible. For an inside deficit-four row, the residual exceptional block has size three. The endpoint branches have no point types; the two nontrivial branches leave 12 and 3 screened factorization cases, all excluded by exact certificates.

Proposition 7.1. *None of the five profiles*

$$6^{16} 7^6, \quad 5^1 6^{14} 7^7, \quad 5^2 6^{12} 7^8, \quad 4^1 6^{13} 7^8, \quad 5^3 6^{10} 7^9$$

can occur in a $K_{3,3}$-free 13×22 matrix with 138 ones.

8. Proof of the main theorem

Proof of Theorem 1.1. The witness in Appendix A proves (2). Suppose a 138-one matrix existed. The penalty enumeration (6) places its degree multiset among 83 profiles. Exact global Farkas certificates exclude 77 profiles. Proposition 4.1 excludes $6^{18} 7^3 9^1$, and Proposition 7.1 excludes the five remaining profiles. Therefore no 138-one matrix exists. Deleting ones shows that no larger $K_{3,3}$-free matrix exists either, so the checked 137-one construction is optimal. □

9. Verification, reproducibility, and trust boundary

The complete proof artifact is designed to separate discovery from verification.

9.1. **One-command verification.** From the repository root, run

```
cd proof
python3 verify_all.py
```

The verifier compiles `local_screen_general.cpp` with a GCC- or Clang-compatible C++17 compiler if needed, then performs the following checks:

1. validates the 137-one witness over all 286 row triples;
2. independently re-enumerates the 83 possible 138-edge degree profiles;
3. replays 77 global rational Farkas certificates;
4. checks the elementary exclusion of $6^{18} 7^3 9^1$;
5. exhaustively regenerates every marked-row point-type and leave case used for the five final profiles;
6. verifies every local and completion certificate coefficient exactly.

The expected final line is

```
ALL EXACT-VALUE VERIFICATION GATES PASSED
```

9.2. **Certificate counts.** The complete development archive also retains an earlier certified 139-edge exclusion. Across all retained gates, the checker verifies 125,983 candidate-block coefficients for the historical 139 proof, 390,039 for the global 138 reduction, and 5,262,392 for the final local proof, totaling

$$5{,}778{,}414$$

exact nonnegativity checks. The 139 component is not logically needed for Theorem 1.1; it is retained as an independent regression test.

9.3. **What is and is not trusted.** The proof depends on:

- exhaustive finite integer enumeration in the supplied C++ source;
- Python integer and `Fraction` arithmetic;
- explicit Farkas vectors stored in JSON;
- the explicit lower-bound witness.

Floating-point linear programming was used during discovery to locate dual vectors. Before inclusion, these vectors were rationalized or integer-scaled, and the final checker verifies their defining inequalities from scratch. No floating-point solver status, MILP infeasibility flag, tolerance, or randomized search result is a premise.

The only symmetry reduction in the integral common-$D = 7$ branch constructs all 8,640 group elements explicitly, verifies that each preserves point types and leave multiplicities, and checks orbit coverage of all 285 labeled systems. Repeated columns remain permitted throughout.

9.4. **Scope and limitations.** The proof is computer-assisted: its load-bearing finite enumerations and certificate replays are carried out by the supplied exact-arithmetic programs. The repository records an adversarial audit of omitted profiles, hidden simplicity assumptions, incomplete leave enumeration, invalid rank or determinant implications, symmetry loss, and numerical tolerance. A second implementation of the local enumerator would provide an additional independent check, but no solver status or floating-point tolerance is used as a premise.

10. Discussion

The proof illustrates a useful division of labor for finite extremal problems. A coarse global relaxation disposes of most degree patterns efficiently. The few profiles surviving the relaxation are not random hard instances: they are nearly balanced around the zero-penalty degrees six and seven, and their small unused triple capacity forces strong congruences on marked-row deficits. Passing to the leave of a marked row converts those congruences into a compact residual packing problem whose Gram matrix is almost determined.

The method is potentially reusable. For nearby $K_{3,3}$-free matrix problems, one may seek a supporting line for $d \mapsto \binom{d}{3}$, enumerate low-penalty degree profiles, apply global dual separation, and reserve exact leave enumeration for the integrality-only residue. The computational burden is then concentrated in small, transparent certificate families rather than a monolithic SAT or MILP run.

Appendix A. The 137-edge witness

Rows are labeled $1, \dots, 13$. The following 22 column supports have total size 137:

$$
\begin{array}{ll}
B_1 = \{2,4,6,7,11,13\} & B_2 = \{3,6,7,8,10,13\} \\
B_3 = \{3,4,6,8,11,12\} & B_4 = \{1,3,8,9,10,12\} \\
B_5 = \{1,3,4,6,9,13\} & B_6 = \{1,2,3,8,11,13\} \\
B_7 = \{1,2,4,6,8,10\} & B_8 = \{1,2,3,6,7,12\} \\
B_9 = \{2,7,8,10,11,12\} & B_{10} = \{2,6,8,9,12,13\} \\
B_{11} = \{2,3,4,7,8,9\} & B_{12} = \{2,3,4,10,12,13\} \\
B_{13} = \{1,2,7,9,10,13\} & B_{14} = \{2,3,6,9,10,11\} \\
B_{15} = \{3,7,9,11,12,13\} & B_{16} = \{1,3,4,7,10,11\} \\
B_{17} = \{1,2,4,9,11,12\} & B_{18} = \{1,5,6,10,11,12,13\} \\
B_{19} = \{1,5,6,7,8,9,11\} & B_{20} = \{4,5,8,9,10,11,13\} \\
B_{21} = \{1,4,5,7,8,12,13\} & B_{22} = \{4,5,6,7,9,10,12\}
\end{array}
$$

The first seventeen blocks have size six and the final five have size seven. The accompanying verifier checks that each row triple occurs in at most two blocks.

Appendix B. Certificate formats

Global and completion certificates store rational multipliers as numerator–denominator pairs. Local factorization certificates are multiplied by a common positive denominator and stored as sparse integer vectors. For each certificate, the verifier reconstructs the complete right-hand side, checks that its dual value is strictly negative, regenerates every eligible block, and checks that the corresponding dual coefficient is nonnegative. Thus certificate files are data, not trusted executable code.

Appendix C. File map

The principal artifact files are:

`proof/verify_all.py:`
Complete verification gate.

`proof/verify_137.py:`
Exhaustive lower-witness check.

`proof/enumerate138.py:`
Independent degree-profile enumeration.

`proof/verify_138_reduction.py:`
Global profile and Farkas replay.

`proof/local_screen_general.cpp:`
Exact point-type and leave-multigraph enumerator.

`proof/verify_138_full.py:`
Local screen, factorization, orbit, and completion replay.

`proof/certs138/:`
Global profile certificates.

`proof/local_certs/:`
Local factorization certificates.

`proof/completion_certs/:`
Global completion certificates for the integral branch.

`proof/ADVERSARIAL_AUDIT_FULL.md:`
Recorded failure-mode audit.

Acknowledgements

The author acknowledges the use of OpenAI GPT 5.6 Sol High in solving the problem and preparing the accompanying verification artifact and manuscript. The author remains responsible for the mathematical content.

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